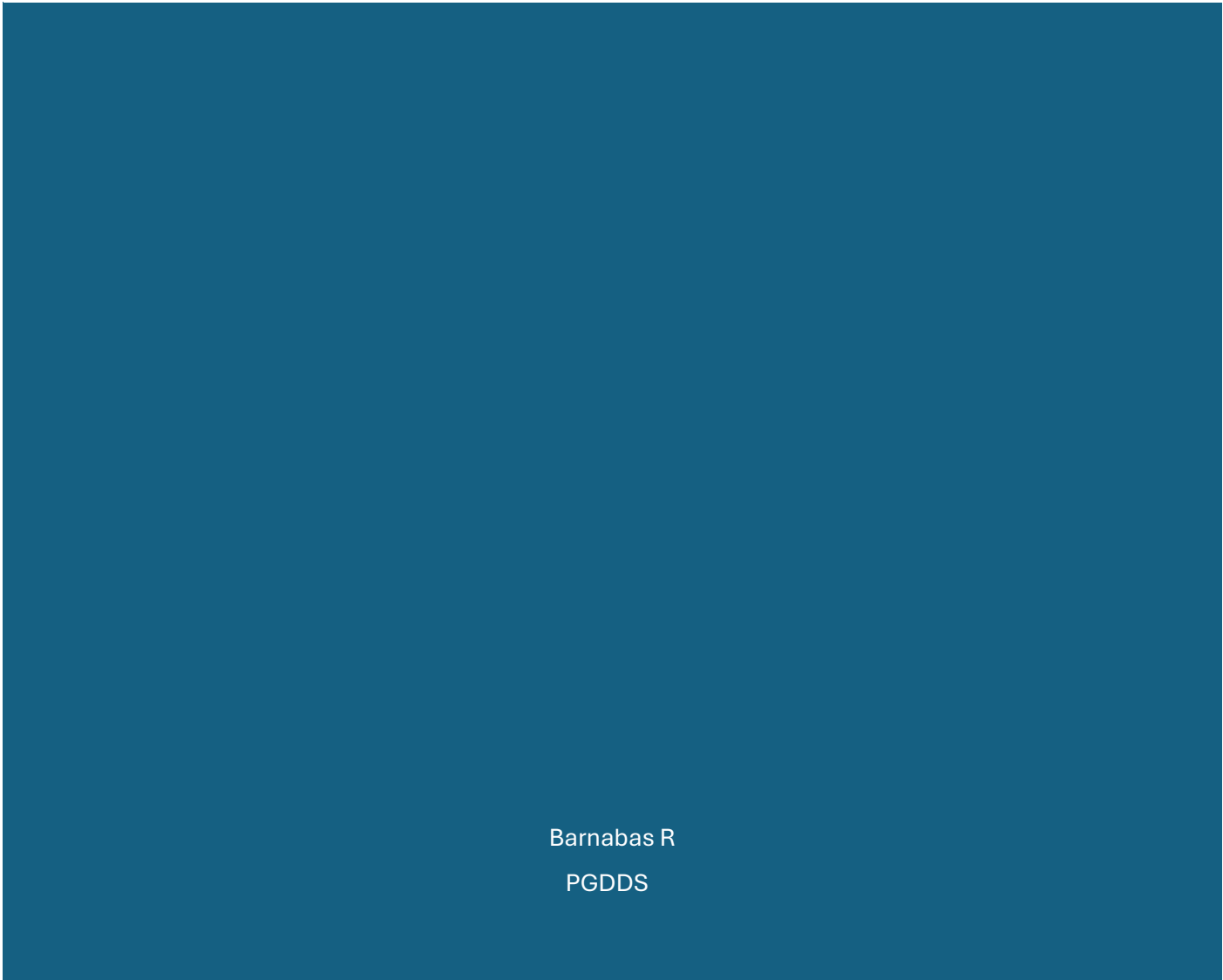




BUSINESS REPORT Capstone Project – Supply Chain Demand Prediction: Final Report



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Capstone Business Report: Supply Chain Demand Prediction

Executive Summary

The primary objective of this project is to forecast the optimal shipment quantity (**product_wg_ton**) per warehouse for an FMCG company to minimize stockouts and excess inventory. The dataset contains **25,000 records** and **24 initial features**. After conducting extensive exploratory data analysis, preprocessing, and advanced modeling, a **Gradient Boosting** model was selected as the final solution, achieving a test **RMSE of 849.29** and an **R-squared of 0.9946**, significantly outperforming the baseline linear model.

1. Understanding the Business Problem

- **Problem Statement:** The company faces operational inefficiencies in inventory distribution, leading to lost sales from stockouts and high carrying costs from waste.
- **Need of the Study:** There is a critical need for a data-driven approach to predict warehouse demand accurately, allowing for optimized logistics and redistribution.
- **Business Opportunity:** Accurate demand forecasting provides an opportunity to implement targeted business actions, such as localized advertising and more precise safety stock adjustments, ultimately improving the bottom line and customer satisfaction.

2. Data Description:

- The dataset contains information about the historical sales patterns. The detailed data dictionary is given below:

- Ware_house_ID: Product warehouse ID
- WH_Manager_ID: Employee ID of warehouse manager
- Location_type: Location of the warehouse, like in a city or a village

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- WH_capacity_size: Storage capacity size of the warehouse
- zone: Zone of the warehouse
- WH_regional_zone: Regional zone of the warehouse under each zone
- num_refill_req_l3m: Number of times refilling has been done in the last 3 months • transport_issue_l1y: Any transport issue, like an accident or goods stolen, reported in the last year • Competitor_in_mkt: Number of instant noodles competitors in the market
- retail_shop_num: Number of retail shops that sell the product under the warehouse area • wh_owner_type: Company owns the warehouse, or they have rented the warehouse • distributor_num: Number of distributors working between the warehouse and retail shops • flood_impacted: Warehouse is in the flood-impacted area indicator
- flood_proof: Warehouse is flood-proof. Like storage is at some height, not directly on the ground
- electric_supply: The Warehouse has an electric backup, like a generator, so they can run the warehouse in load shedding
- dist_from_hub: Distance between the warehouse to the production hub in Kms • workers_num: Number of workers working in the warehouse
- wh_est_year: Warehouse established year
- storage_issue_reported_l3m: Warehouse reported a storage issue to the corporate office in the last 3 months. Like rats, fungus due to moisture, etc.
- temp_reg_mach: Warehouse has a temperature regulating machine indicator
- approved_wh_govt_certificate: What kind of standard certificate has been issued to the warehouse from the government regulatory body
- wh_breakdown_l3m: Number of times the warehouse faced a breakdown in the last 3 months. Like a strike by workers, a flood, or an electrical failure
- govt_check_l3m: Number of times government officers have visited the warehouse to check the quality and expiry of stored food in the last 3 months
- product_wg_ton: Product has been shipped in the last 3 months. Weight is in tons

3. Exploratory Data Analysis (EDA)

- **Data Overview:** The dataset consists of 25,000 observations. Key features include warehouse location, management type, operational issues, and regional zones.
- The dataset contains a mix of categorical (e.g., Location_type, zone) and numerical variables (e.g., retail_shop_num, workers_num). Initial checks confirm there are no duplicate rows in the dataset.

Univariate Analysis:

Target Variable (product_wg_ton):

The average shipment weight is approximately 22,102.63 tons, with a range between 2,065 and 55,151 tons. The distribution appears multi-modal, suggesting different tiers of warehouse demand.

Retail Shop Count:

The distribution for retail_shop_num (Mean: 4,985) is right-skewed (Skewness: 0.908), suggesting a majority of warehouses serve a standard number of shops, while a few serve significantly larger markets.

Hub Distance:

dist_from_hub (Mean: 163.5 km) shows a uniform distribution, indicating warehouses are evenly spread across various distances from central hubs.

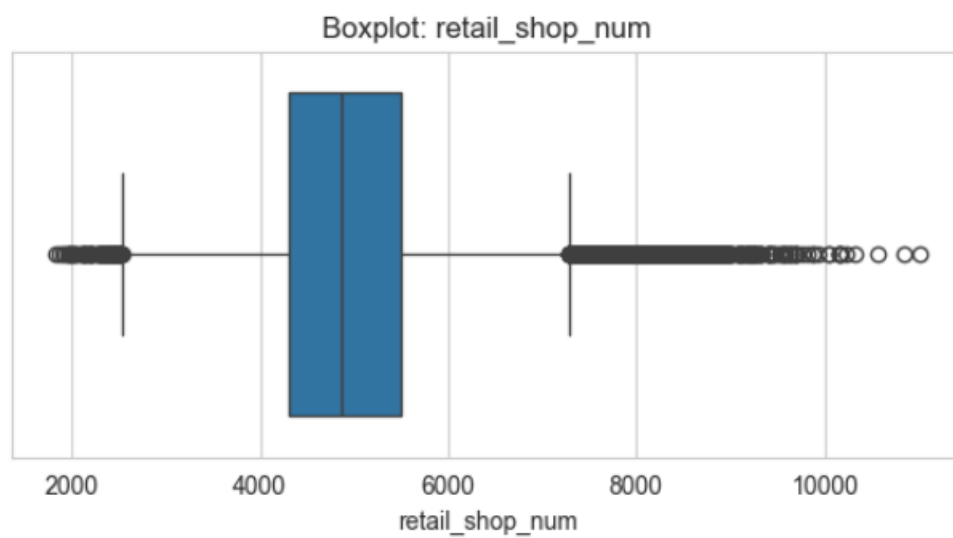
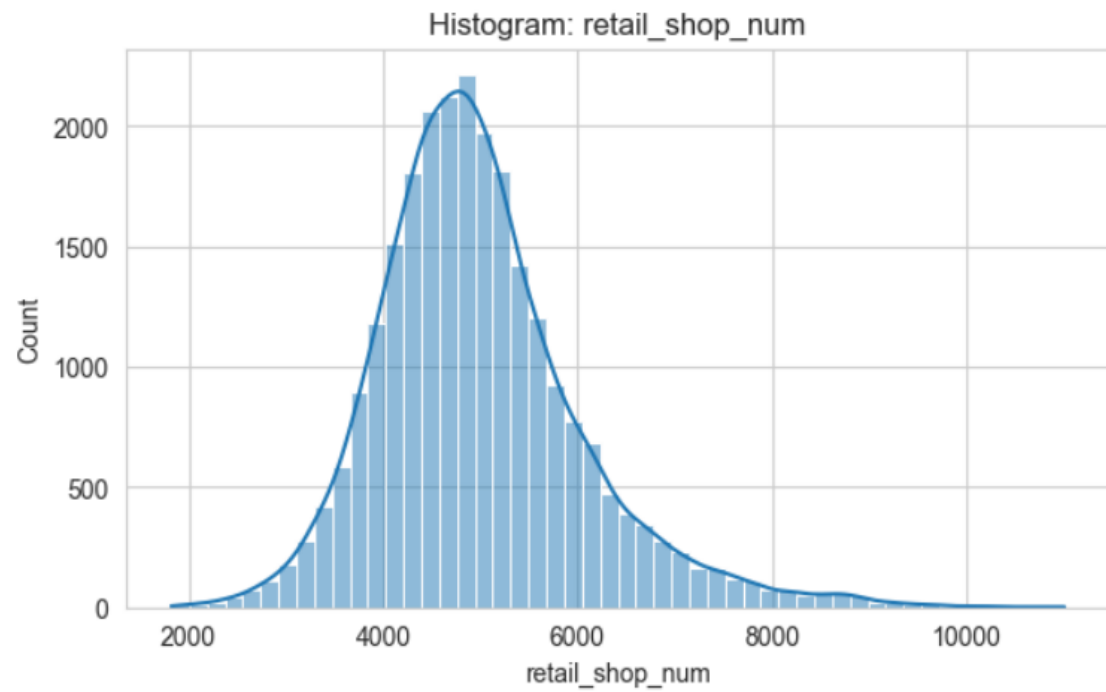


Figure 1 Histogram and box plot of count of retail shops

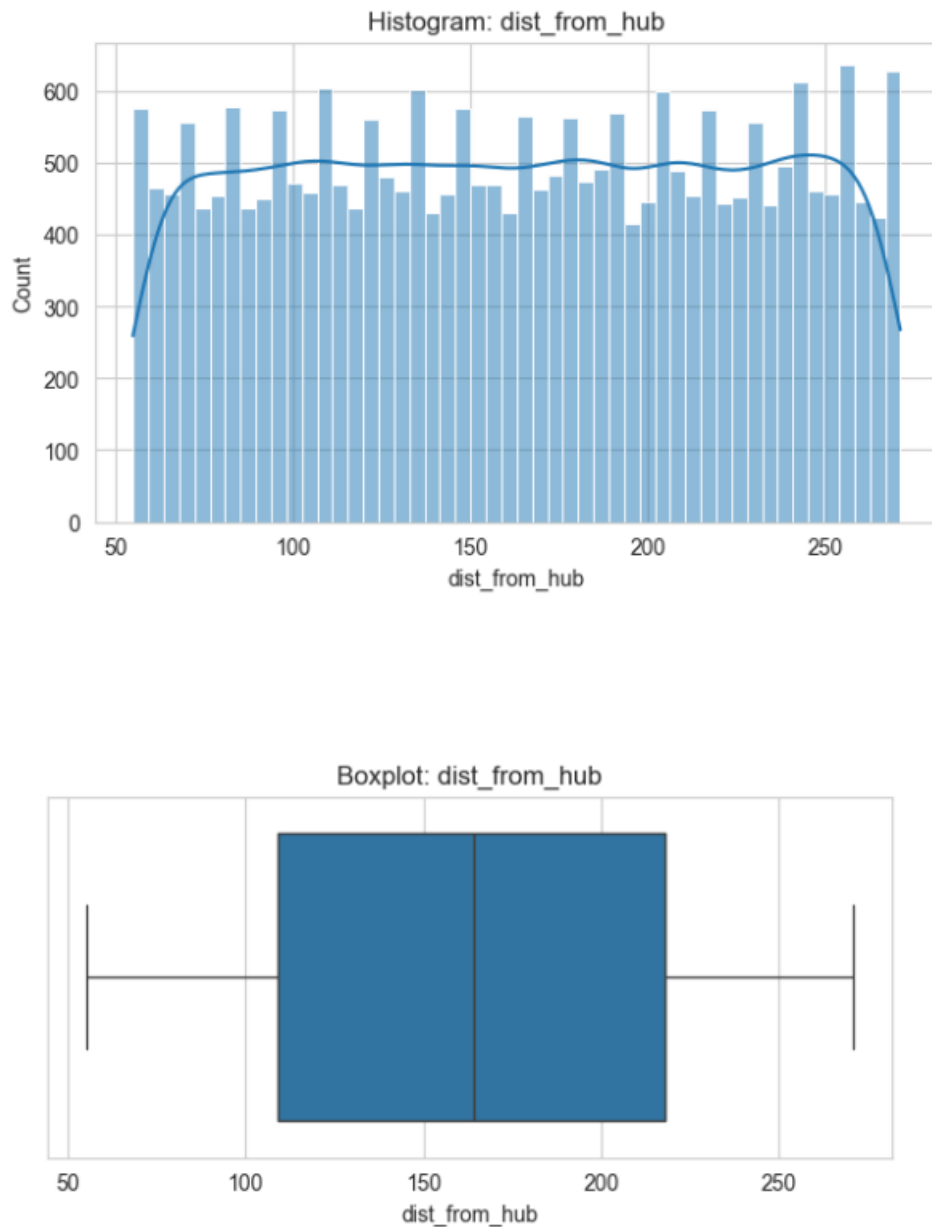


Figure 2 Distribution and Box Plot of distance from hub

Target Variable Analysis:

The target variable, `product_wg_ton`, has a mean of 22,102.63 tons. The distribution is multimodal, with notable peaks around the 7,000 and 30,000-ton marks, indicating different tiers of warehouse capacity or regional demand.

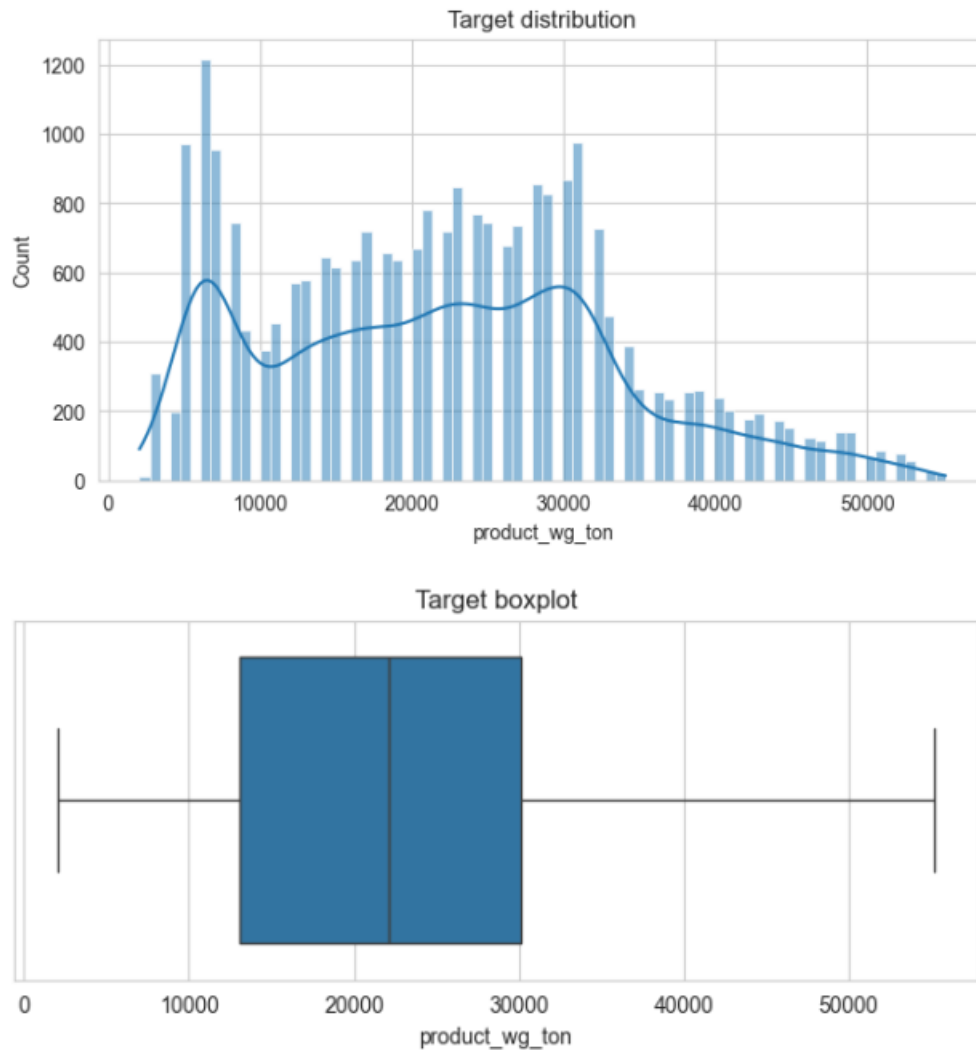
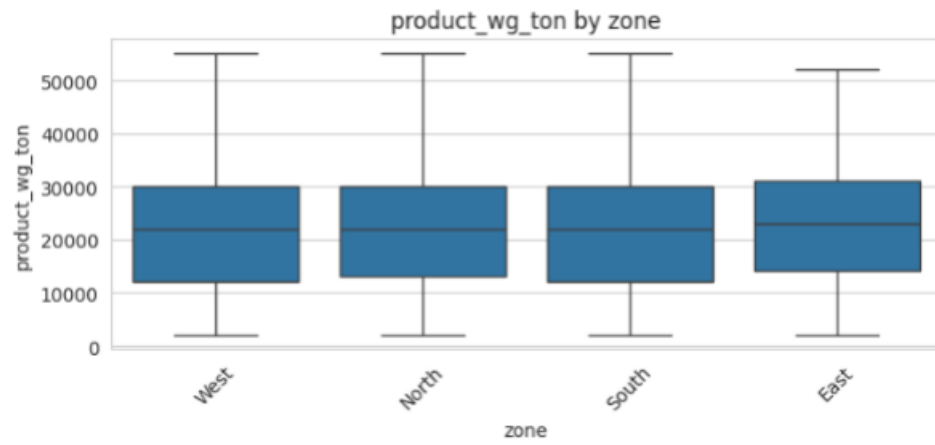


Figure 3 Detailed Distribution of the Target Variable (*product_wg_ton*)

Bivariate Analysis: Key Drivers of Demand

- **Location Type:** Urban warehouses handle a higher mean tonnage (~25,000) compared to Rural warehouses (~22,000).
- **Regional Variance:** The East zone shows the highest mean shipment weight (22,721 tons), while the South zone is the lowest (21,933 tons).
- **Warehouse Age:** There is a clear negative trend between *wh_est_year* and demand.

Older warehouses (those established in earlier years) tend to manage significantly higher tonnages.



	count	mean	median	std
WH_regional_zone				
Zone 2	2963	22,470.728	23,061.000	11,882.148
Zone 4	4176	22,173.379	22,121.000	11,644.451
Zone 6	8339	22,115.560	22,099.000	11,496.958
Zone 5	4587	22,022.588	22,075.000	11,773.081
Zone 3	2881	21,968.146	22,112.000	11,421.767
Zone 1	2054	21,742.713	21,129.500	11,464.235

Figure 4 Average product weight by warehouse zone

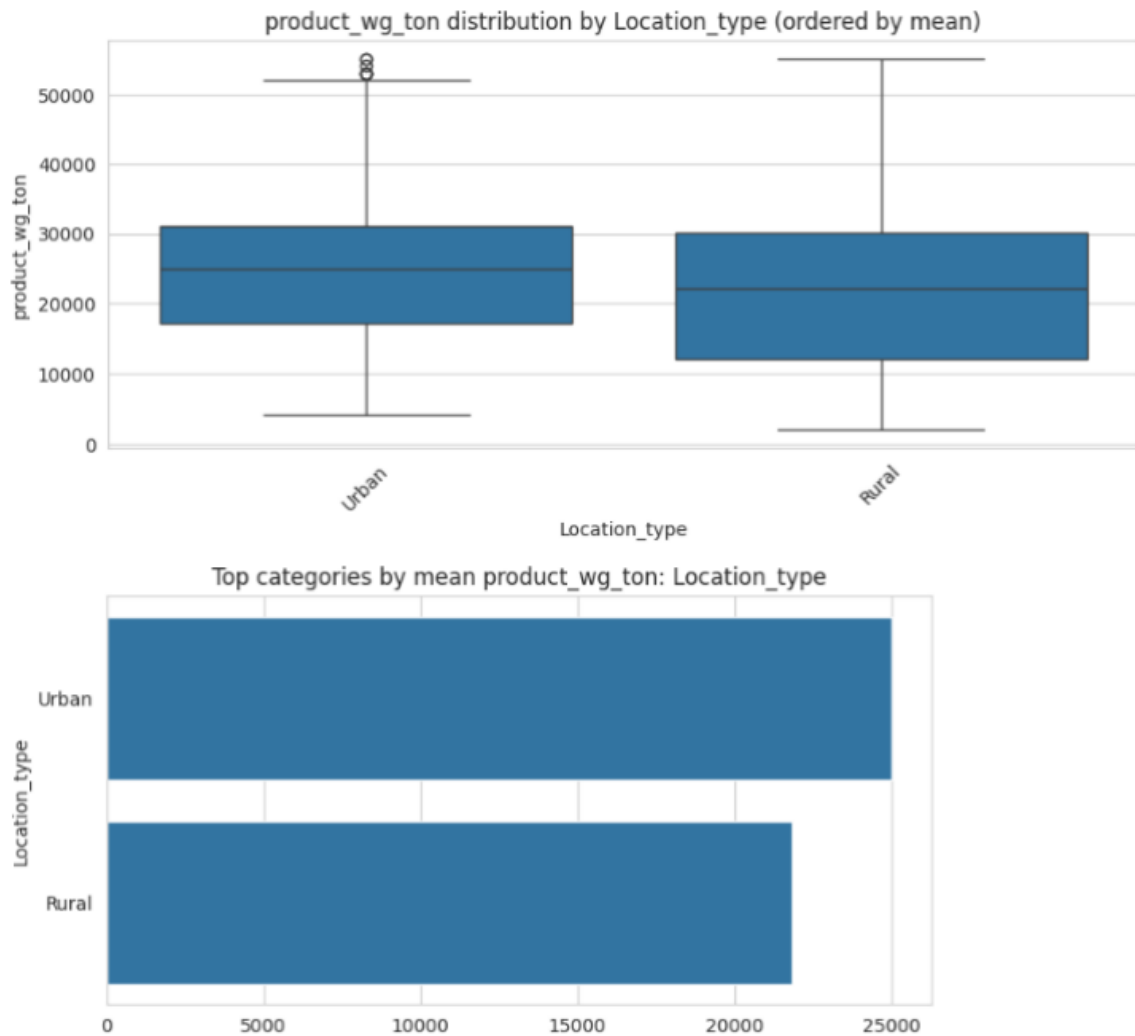


Figure 5 Impact of Location Type (Urban vs. Rural) on Shipment Weight

Multivariate Analysis:

The correlation analysis reveals that storage_issue_reported_l3m has a near-perfect positive correlation (0.99) with the target, while wh_est_year has a strong negative correlation (-0.83).

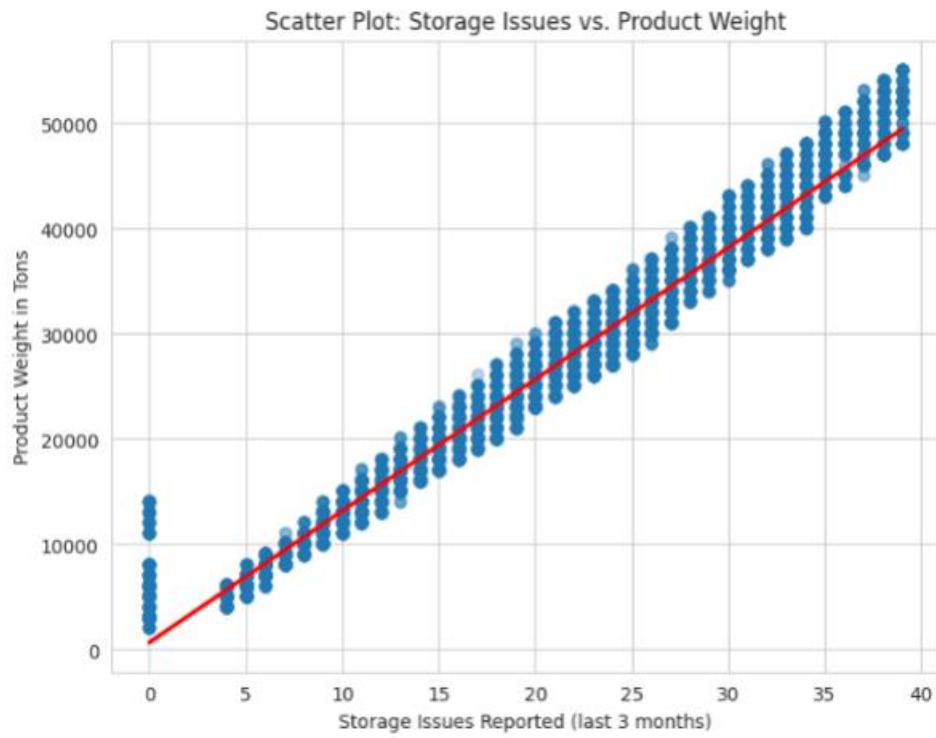


Figure 6 Scatter Plot of Storage Issues vs. Product Weight

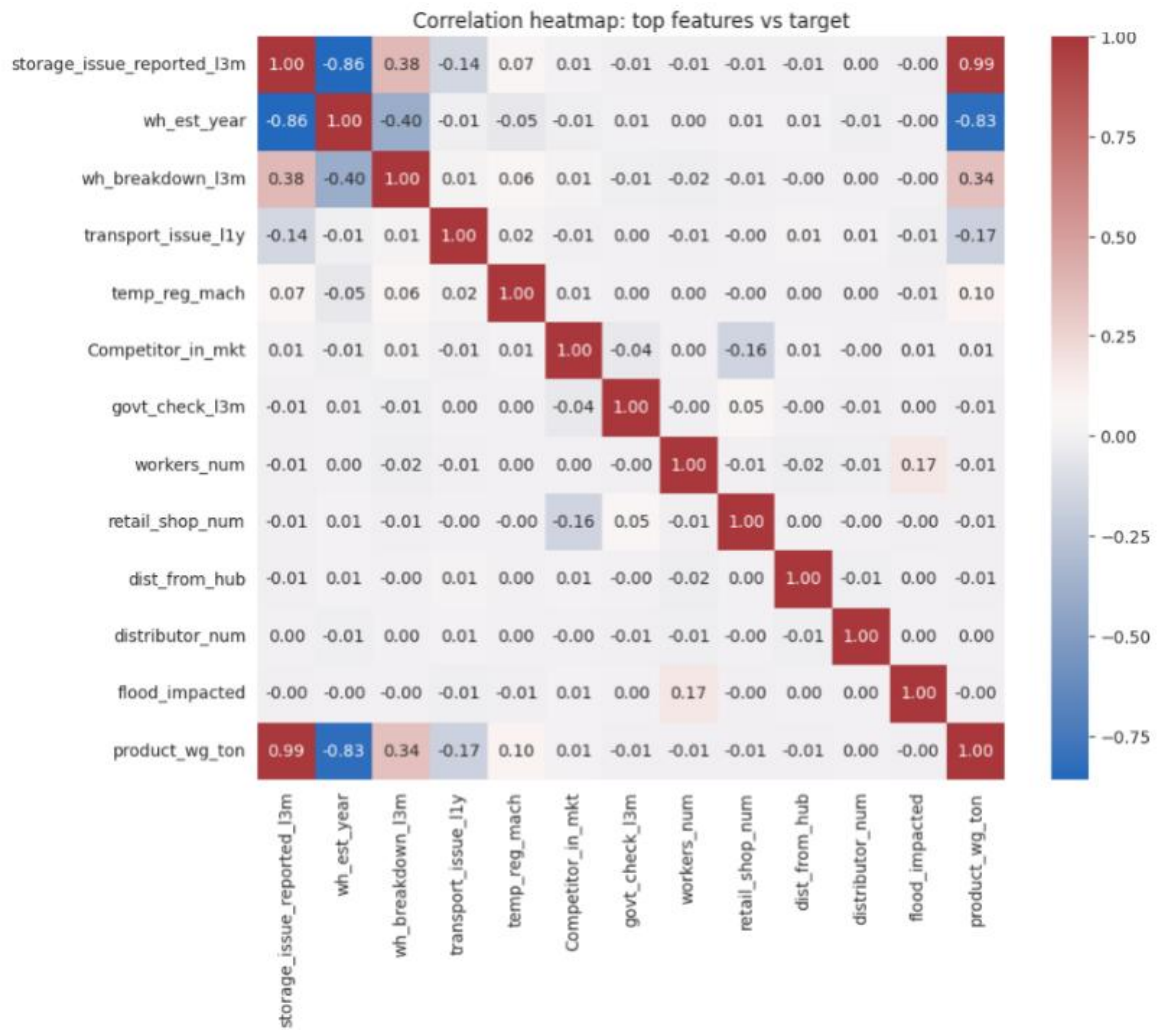


Figure 7 Correlation Heatmap: Top Features vs. Target

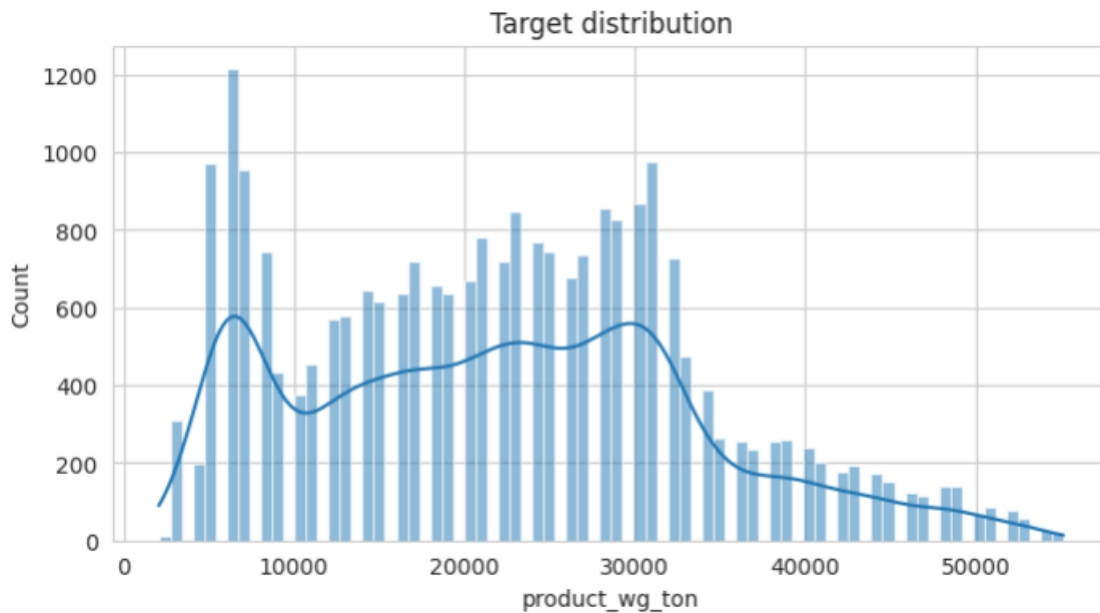


Figure 8 Distribution of Target Variable (Product_wg_ton)

4. Data Preprocessing

Missing Value Identification:

A significant portion of the `wh_est_year` data is missing (47.52%). Other columns with missing values include `workers_num` (3.96%) and `approved_wh_govt_certificate` (3.63%). Analysis shows that missingness in `approved_wh_govt_certificate` correlates with the target variable (-0.279), suggesting the data is not missing at random.

Outlier Detection Extreme outliers were identified using the Interquartile Range (IQR) method:

- `transport_issue_11y`: 348 extreme outliers.

- `flood_impacted`: 2,454 extreme outliers.

- `product_wg_ton`: The 99.9th percentile is 54,130 tons.

Feature Engineering:

Several business-relevant features were proposed to enhance model performance:

- `wh_age`: Calculated as 2026 minus the establishment year.

- `risk_index`: A composite score summing binary indicators for transport issues, flood impact, and storage breakdowns.

- `refills_per_month`: Derived from the three-month refill request count.

Data Leakage Check:

Initial scans for data leakage identified that `retail_shop_num` had one instance where the value exactly equaled the target variable.

The analysis employs two primary "leakage heuristics" to flag suspicious features:

- **Exact Equality Check:** The code checks if any feature value exactly equals the target variable (`product_wg_ton`). In the source data, this identified a specific warning for **`retail_shop_num`**, which had one instance of exact equality with the target.
- **High Correlation Threshold:** Features are screened for an **extremely high correlation (absolute value > 0.995)** with the target variable, which often indicates that the feature contains information not typically available at the time of prediction.

Feature Exclusion

1. To prevent leakage from identifiers or non-predictive metadata, specific columns are explicitly excluded from the model's feature set:
2. **Identifier Columns:** Features such as **`Ware_house_ID`** and **`WH_Manager_ID`** are defined as identifiers and removed before training to ensure the model does not "memorize" specific entities rather than learning general patterns.
3. **Strategic Feature Engineering**
4. The sources highlight potential leakage risks during feature creation. For example, a feature called **`cap_util_proxy`** was calculated by dividing the target (**`product_wg_ton`**) by the warehouse capacity. In the preprocessing checks, this feature showed **100% missingness or was flagged** because using the target to create a training feature is a primary source of data leakage
5. **Preparation:** Features were scaled using **`StandardScaler`**, and categorical variables were encoded using **`OneHotEncoder`** within a transformation pipeline

5. Data Preprocessing

- **Missing Values:** Significant missingness was found in `wh_est_year` (**47.52%**). Missingness in government certificates and establishment years was found to correlate with the target, indicating "informative missingness".
- **Outlier Detection:** Using the 3xIQR method, extreme outliers were identified in `flood_impacted` (2,454 cases) and `transport_issue_l1y` (348 cases).
- **Feature Engineering:** New features were created to capture business logic:
 - **wh_age:** Calculated as 2026 minus establishment year.
 - **risk_index:** A composite score of transport, flood, and breakdown issues.
 - **refills_per_month:** Normalizing refill requests over time.

6. Model Building – Baseline Model

- **Metric of Choice: Root Mean Squared Error (RMSE)** was selected to penalize large forecasting errors, alongside **R-squared (R2)** to measure explained variance.
- **Baseline Model:** A **Linear Regression** model served as the benchmark.
- **Performance:** The baseline achieved an **RMSE of 1,127.53** and an **R2 of 0.9905**. While high, the error magnitude (MAE: 874.94) left room for improvement through non-linear modeling.

7. Model Building – Advanced Models

Five advanced ensemble and tree-based models were trained to capture non-linear relationships:

1. **Decision Tree:** RMSE 1,233.87
2. **Bagging Regressor:** RMSE 884.74

3. **Random Forest:** RMSE 881.38
4. **AdaBoost:** RMSE 1,760.45
5. **Gradient Boosting:** RMSE 859.84

8. Model Performance Improvement: Hyperparameter Tuning

The three most promising models (**Random Forest, Gradient Boosting, and AdaBoost**) were selected for optimization using RandomizedSearchCV.

- **Reasoning:** Gradient Boosting and Random Forest showed the lowest initial errors, while AdaBoost was tuned to see if its performance could be stabilized.

Results: Tuning reduced the RMSE for Gradient Boosting to **849.29** and for Random Forest to **874.20**

9. Model Performance Comparison and Final Model Selection

- **Final Selection: Gradient Boosting** was selected as the final model due to having the lowest test RMSE and highest R-squared.
- **Final Test Metrics:**
 - **RMSE:** 849.29
 - **MAE:** 635.66
 - **R2:** 0.9946
 - **MAPE:** 4.00%

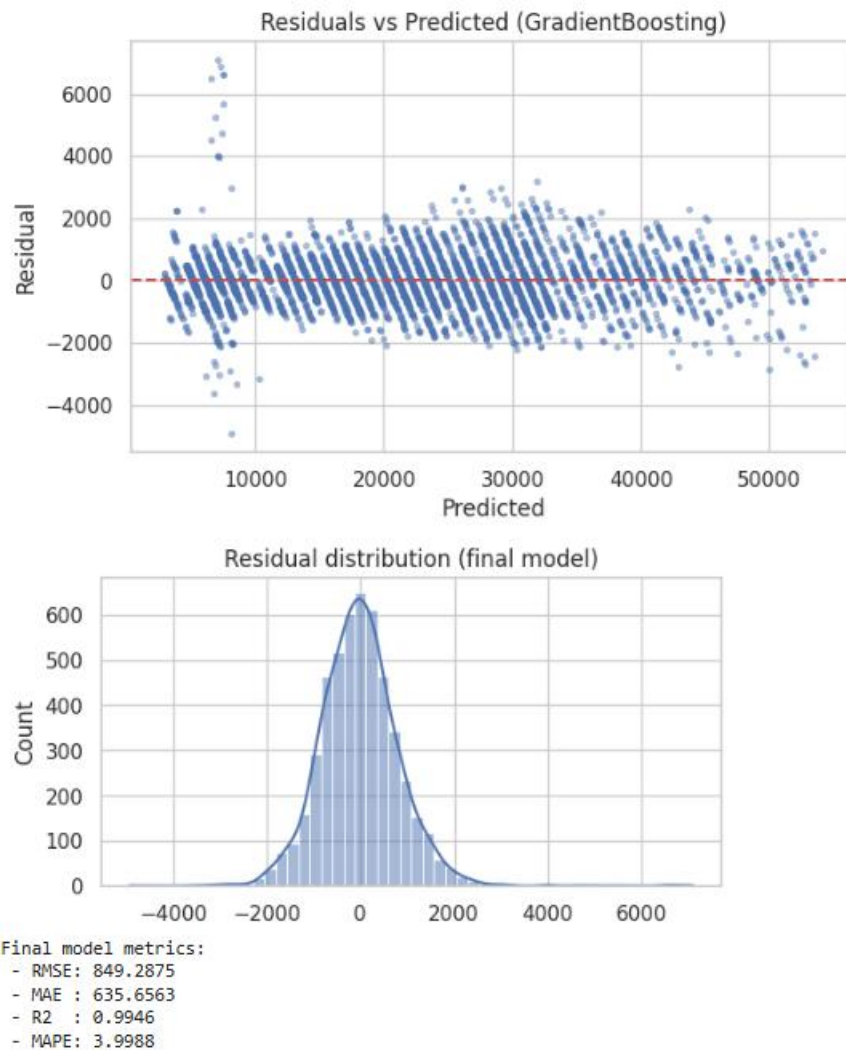


Figure 9 Residuals vs. Predicted Values for Final Model

10. Actionable Insights & Recommendations

1. **Inventory Prioritization:** High-demand warehouses identified in the EDA (e.g., those in the North zone) should receive priority for shipment scheduling to prevent stockouts.
2. **Dynamic Safety Buffers:** Use the **risk_index** feature to add a safety stock buffer to predictions; warehouses with higher flood or transport risks require 10-15% more inventory overhead.

3. **Targeted Maintenance:** Older warehouses (wh_age) handle higher volumes but are more prone to breakdowns. Prioritize these facilities for infrastructure resilience investments.
4. **Logistics Rebalancing:** With dist_from_hub being uniform, the company should re-evaluate distribution routes for long-distance warehouses to reduce lead times.
5. **Leading Indicators:** Use **refills_per_month** and **retail_shop_num** as primary operational triggers for demand planning, as these were identified as the most critical features in model decision-making.